C4: the interval predicate is pseudo-generic (conjecture plus toy energy measurement)

Lead C4 of `RESEARCH_PRIME_FIELD_LEADS_20261031.md`; thresholds frozen in `../PLAN.md`. Counted metric: curve additions (72,367,912 in total). No wall-time claim.

1. What this is not

The lab card `LEAD-EXO-dl-character-sum-pseudorandom` (H-PSEUDO, `exhausted_at_scale`) measured the additive-character sum of the **discrete log** over the small-x base, `sum_{P in F} e(2 pi i k DL(P)/N)`, against a matched null (ratio flat at 1.28-1.34 over $2^{20..2}26$). C4 measures a different object: the additive **energy of the base itself inside the group**,

```
E(F) = #{(P1,P2,P3,P4) in F^4 : P1 + P2 = P3 + P4}    (ordered, trivial solutions included),
```

which needs no discrete log (it is computed by hashing the m^2 pair sums). The two are related through Fourier duality over the cyclic group (E(F) = (1/N) sum_k |hat 1_F(k)|^4), so a flat character sum bounds energy from above, but the card's measured constant 1.28-1.34 is a constant factor in the *sup norm* and says nothing about the 4th moment at the 1 + o(1) level. C4 asks the 4th-moment question directly.

2. Conjecture (precise form)

Let E/F_p have prime order N (cyclic group). For B in [1,p] let F_B = {P in E(F_p) {O} : x(P) < B} (closed under negation). Let R_m be a uniformly random **negation-closed** subset of E(F_p){O} with |R_m| = |F_B| = m.

Conjecture (pseudo-genericity of the interval predicate). For every fixed delta > 0, uniformly over curves E over F_p, B with m = |F_B| >= N^delta (and m <= N^{1-delta}),

```
E(F_B) / E[E(R_m)] = 1 + o(1)    as p -> infinity,
```

with E[E(R_m)] = 3m^2 + m^4/N (1 + o(1)) + O(m) (trivial classes P1=P3,P2=P4 ; P1=P4,P2=P3 ; P1=-P2,P3=-P4, plus the generic term m^4/N).

Remark on the null. Against a *plain* uniform random subset the ratio is not 1 even in theory: negation closure adds the third trivial class, so the plain-null ratio tends to (3m^2 + $m^{4/N)/(2m}$2 + m^4/N), i.e. 4/3 at m = sqrt(N) and 3/2 at m = N^{1/3}. This is exact bookkeeping and does not depend on x < B. The conjecture is therefore stated against the negation-closed null (design choice made before running; the plain-null ratio is also recorded below).

Why it would be the missing yield lemma of L1. A sieve-style relation shortcut on the interval base needs the base to have more additive coincidences than a random set of its density (excess energy = excess relations among m^2 pair sums, i.e. super-random decomposition yield; compare the Betti-yield duality KN-FIND-9d2f56 cited by the H-PSEUDO card). The conjecture says the coincidence count of F_B equals the random one to 1 + o(1); by the density/self-reducibility ceiling this would leave the sieve with exactly the generic yield L1 prices. It closes the predicate-level shortcut only for energy (4th moment); higher moments (8-fold coincidences) and non-additive

algebraic structure (Semaev summation polynomials) are not covered. Nothing here proves the conjecture.

3. Literature (named with the confidence stated)

- Shparlinski and coauthors have a body of work on the distribution and exponential sums of points of elliptic curves, including points with coordinates in short intervals/boxes, and on sums of x-coordinates (e.g. Kohel-Shparlinski, "On exponential sums and group generators for elliptic curves over finite fields", ANTS 2000 - confident of that title; the box-distribution papers I describe but do not cite by title, to avoid fabrication). These give *equidistribution* of x in intervals (counts of points per box to p^{1/2+o(1)} error) - a first-moment statement. They do not, to my knowledge, give the 4th-moment / energy statement here at m ~ N^{1/2}.
- Additive-combinatorial background: additive energy, Balog-Szemeredi-Gowers; sum-product style results in elliptic-curve groups. Only used as the framework; no specific theorem is invoked.
- The H-PSEUDO card (KN-FIND-e7a3b1) records that six standard analytic proof routes to the related character-sum statement are closed; C4 inherits that obstruction for a proof and offers only evidence.

4. Toy measurement

Program `energy.rs` (standalone, `rustc -O`, own splitmix RNG, affine curve addition over u64; no crate changes). For k = 12..18, p = largest prime below 2^k (7 sizes); 8 random Weierstrass curves per p with prime non-anomalous order N (candidates with N composite or N = p are rejected; tries per p: 96, 123, 108, 152, 88, 67, 183). Per curve and size m in {round(sqrt N), round(N^{1/3})} (rounded up to even so F_B is negation-closed): F_B = first m points ordered by x; E(F_B) exact by hashing/sorting all m^2 pair sums; 8 uniform random m-subsets and 8 random negation-closed m-subsets on the same curve. Reported: ratio of E(F_B) to the mean of the 8 negation-closed nulls (primary) and to the mean of the 8 plain-uniform nulls (frozen-text version). Sizes reach m = 512, N ~ 2.6e5. Files: `data_per_curve.csv` (every curve, every cell; no cell failed or timed out), `data_summary.csv`, `graph_energy_ratio.svg/.pdf`, `slope_check.txt`.

Observed (mean over 8 curves, sd over curves in parentheses):

log2 p	sqrt N: ratio vs closed null	cbrt N: ratio vs closed null	sqrt N: vs plain	cbrt N: vs plain
12	1.0038 (0.0178)	0.9943 (0.0255)	1.317	1.436
13	1.0014 (0.0173)	1.0126 (0.0176)	1.330	1.471
14	0.9927 (0.0080)	1.0012 (0.0144)	1.317	1.461
15	0.9973 (0.0119)	1.0032 (0.0112)	1.328	1.468
16	0.9999 (0.0053)	1.0052 (0.0091)	1.331	1.481
17	0.9995 (0.0045)	1.0011 (0.0065)	1.330	1.482
18	1.0023 (0.0029)	1.0002 (0.0058)	1.335	1.483

- Largest |mean ratio - 1| is 0.0073 (sqrt N, p=2^14) and 0.0126 (cbrt N, p=2^13), each below 1 sd over curves (0.91 and 0.71 of the curve sd at their worst point); least-squares slope of the mean ratio against log2 p: −3.8e-5 (sqrt N) and −4.9e-5 (cbrt N) per bit, i.e. no growth.
- Within-curve relative sd of the null energies falls from 0.024 to 0.004 across the range (`null_rel_sd_within_curve`), the same scaling as the spread of the ratio over curves.
- Versus the plain null the ratio sits at 1.32-1.34 (sqrt N) and 1.44-1.48 (cbrt N), approaching the exact 4/3 and 3/2 predicted for negation closure alone (Section 2). It does not grow beyond that.
- Binomial/Poisson reference: the generic term lambda = m^4/N is about 262,000 at sqrt N, p~2^18 (about 25% of E) and about 64 at cbrt N (about 0.5% of E). The observed within-curve relative sd of the 8 negation-closed nulls at p~2^18 is 0.0038 (`null_rel_sd_within_curve`); no separate Poisson-variance computation was made (pair sums are clustered, so Poisson would understate it).

5. Decision (frozen rule)

A ratio departing from 1 by more than the spread AND growing with p: not observed against the matched (negation-closed) null, in either size. Against the plain null the departure is large but constant and is the exact negation-closure effect. Decision: **null_confirmed** at toy scale.

6. Limits

- At m = N^{1/3} the generic term m^4/N (about 64 at N~2^18) is dwarfed by the trivial 3m^2 (about 12,000); the cbrt N energy ratio is dominated by trivial classes and is nearly blind to non-trivial structure. The separate "non-trivial part" ratio was undefined (negative or noise-level after subtracting) there, hence NaN in `data_summary.csv`. Only the sqrt N column carries real power, and even there the signal is the m^4/N term, about 25% of E at m = sqrt N.
- Range $2^{12..2}18$ only; the card's own revisit condition (growth beyond 2^30) is not touched.
- The test sees the 4th moment only. Structure in the Semaev polynomial system or higher-order coincidences is not measured.
- The closure formula 3m^2 - 2m for trivial solutions is approximate (overlaps); the primary ratio does not use it, only the diagnostic non-trivial column does.

Reading

Energy of the interval base matches a random negation-closed set of the same size to within 1% at every tested size and shows no trend in p, which is what the conjecture predicts; this is evidence consistent with the conjecture at toy scale, not a proof, and says nothing about higher moments or algebraic structure.